

1. Write a program to print whether a number is even or odd, also take input from the user.

```
class Prog1
{
    public static void main(String[] args)
    {
        Scanner scn=new Scanner(System.in);
        System.out.println("ENTER NUM1");
        int num1=scn.nextInt();
        display(num1);
    }
    public static void display(int num1)
    {
        if(num1%2==0)
        {
            System.out.println("IT IS EVEN NUMBER");
        }
        else
        {
            System.out.println("IT IS ODD NUMBER");
        }
    }
}
```

OUTPUT

```
ENTER NUM1
123
IT IS ODD NUMBER
```

```
ENTER NUM1
2468
IT IS EVEN NUMBER
```

CONCEPT

Even Number

A number is called **even** if it is completely divisible by 2.

That means:

- remainder = 0
- divisible by 2

Examples:

2, 4, 6, 8, 10, 12

Odd Number

A number is called **odd** if it is **not completely divisible by 2**.

That means:

- remainder $\neq 0$
- usually remainder = 1

Examples:

1, 3, 5, 7, 9, 11

How to identify in programming?

We use **modulus operator %**.

Syntax:

number % 2

This gives remainder after division by 2.

Case 1: Even

$$8 \% 2 = 0$$

Remainder is 0 $\rightarrow$ Even

Case 2: Odd

$$9 \% 2 = 1$$

Remainder is 1 $\rightarrow$ Odd

2.Take 2 numbers as input and print the largest number.

```
import java.util.Scanner;
```

```
class Prog2
```

```

{
    public static void main(String[] args)
    {
        Scanner scn=new Scanner(System.in);
        System.out.println("ENTER NUM1");
        int num1=scn.nextInt();
        System.out.println("ENTER NUM2");
        int num2=scn.nextInt();
        display(num1,num2);
    }
    public static void display(int num1,int num2)
    {
        if(num1>num2)
        {
            System.out.println("NUM1 IS LARGEST THEN NUM2");
        }
        else
        {
            System.out.println("NUM2 IS LARGEST THEN NUM1");
        }
    }
}

```

OUTPUT

```

ENTER NUM1
348855
ENTER NUM2
1234797
NUM2 IS LARGEST THEN NUM1

```

CONCEPT

Largest Number

To find the largest number between two numbers:

- Compare both numbers using >
- If first number is greater → first is largest

- Otherwise → second is largest

Example:

- 10 and 5 → 10 is largest
- 7 and 12 → 12 is largest

3.To calculate Fibonacci Series up to n numbers.

```
import java.util.Scanner;

class Prog3
{
    public static void main(String[] args)
    {Scanner scn=new Scanner(System.in);
        System.out.println("ENTER NUM");
        int num=scn.nextInt();
        int n1=0;
        int n2=1;
        int n3=n1+n2;
        display(n1,n2,n3,num);
    }
    public static void display(int n1,int n2,int n3,int num)
    {
        for(int i=1;i<=num;i++)
        {
            System.out.println(n1);
            n1=n2;
            n2=n3;
            n3=n1+n2;
        }
    }
}
```

OUTPUT

```
ENTER NUM
10
0
1
1
2
3
5
8
13
21
34
```

CONCEPT

Fibonacci Concept

Fibonacci series is a sequence where each number is the sum of previous two numbers.

Formula:

next = first + second

Series:

0, 1, 1, 2, 3, 5, 8, 13, 21... How it works

4.To find out whether the given String is Palindrome or not.

```
class Prog4
```

```
{
    public static void main(String[] args)
    {
        Scanner scn=new Scanner(System.in);
        System.out.println("ENTER STRING");
        String s1=scn.next();
        String rev="";
        char ch=' ';
        display(s1,rev,ch);
    }
    public static void display(String s1,String rev,char ch)
    { for(int i=s1.length()-1;i>=0;i--)
        {
            ch =s1.charAt(i);
            rev=rev+ch;
        }
    }
}
```

```

        System.out.println(rev);

        if(rev.equals(s1))
        {
            System.out.println("IT IS PALINDROM");
        }
        else
        {
            System.out.println("IT IS NOT PALINDROM");
        }
    }
}

```

OUTPUT

```

ENTER STRING
genc
cneg
IT IS NOT PALINDROM

```

```

ENTER STRING
refer
refer
IT IS PALINDROM

```

CONCEPT

Palindrome Concept

A string is called palindrome if it reads same forward and backward.

Examples:

madam

level

Malayalam

5. Input currency in rupees and output in USD.

```

import java.util.Scanner;

class Prog6
{
    public static void main(String[] args)
    {
        Scanner scn=new Scanner(System.in);
    }
}

```

```

        System.out.println("ENTER RUPEES");
        int num1=scn.nextInt();
        double USD=num1/90.20;
        System.out.println("USD :"+USD);
    }
}

```

OUTPUT

```

ENTER RUPEES
54000
USD :598.6696230598669

```

6.Take in two numbers and an operator (+, -, *, /) and calculate the value.

```

import java.util.Scanner;
class Prog5
{
    public static void main(String[] args)
    {
        Scanner scn=new Scanner(System.in);

        System.out.println("ENTER NUM1");
        int num1=scn.nextInt();
        System.out.println("ENTER NUM2");
        int num2=scn.nextInt();
        System.out.println("1.add\n2.sub\n3.mul\n4.div");
        System.out.println("ENTER OPTION");
        int opt=scn.nextInt();
        display(num1,num2,opt);
    }
    public static void display(int num1,int num2,int opt)
    {
        switch(opt)

```

```

{
    case 1:

        int a=num1+num2;
        System.out.println(a);
        break;
    case 2:

        int b=num1-num2;
        System.out.println(b);
        break;
    case 3:

        int c=num1*num2;
        System.out.println(c);
        break;
    case 4:

        int d=num1/num2;
        System.out.println(d);
        break;
    default:

        System.out.println("NO INPUTS");

}
}
}

```

OUTPUT

ENTER NUM1	ENTER NUM1
1234	34
ENTER NUM2	ENTER NUM2
684234	2234
1.add	1.add
2.sub	2.sub
3.mul	3.mul
4.div	4.div
ENTER OPTION	ENTER OPTION
1	3
685468	75956

7. Write a program to input principal, time, and rate (P, T, R) from the user and find Simple Interest.

```
import java.util.Scanner;

class Prog7
{
    public static void main(String[] args)
    {
        Scanner scn=new Scanner(System.in);
        System.out.println("ENTER PRINCIPAL AMOUNT");
        int p=scn.nextInt();
        System.out.println("ENTER YEAR");
        int t=scn.nextInt();
        System.out.println("ENTER RATE OF INTEREST");
        int r=scn.nextInt();
        int SI=p*t*r/100;
        System.out.println("SIMPLE INTEREST AMOUNT :"+SI);
    }
}
```

OUTPUT

```
ENTER PRINCIPAL AMOUNT
6000
ENTER YEAR
1
ENTER RATE OF INTEREST
4
SIMPLE INTEREST AMOUNT :240
```

CONCEPT

Simple Interest is the interest calculated only on the original principal amount.

Formula

$$SI = (P \times T \times R) / 100$$

Where:

- **P** = Principal amount (initial money)

- **T** = Time (years)
- **R** = Rate of interest (%)

8. Take name as input and print a greeting message for that particular name.

```
import java.util.Scanner;

class Prog8 {

    public static void main(String[] args) {

        Scanner scn = new Scanner(System.in);

        System.out.println("Enter your name:");
        String name = scn.nextLine();

        System.out.println("Hello " + name + "! Welcome!");
    }
}
```

OUTPUT

```
Enter your name:
BOB
Hello BOB! Welcome!
```

9. To find Armstrong Number between two given number.

```
import java.util.Scanner;

class Prog9
{
    public static void main(String[] args)
    {
        Scanner scn=new Scanner(System.in);
        System.out.println("ENTER START");
        int start=scn.nextInt();
        System.out.println("ENTER END");
        int end=scn.nextInt();
```

```

        for(int i=start;i<=end;i++)
        {
            int num =i;
            int sum=0;
            int dis=String.valueOf(i).length();

            while(num>0)
            {
                int rem=num%10;
                sum+=(int)Math.pow(rem,dis);
                num=num/10;
            }
            if(sum==i)
            {
                System.out.println(i);
            }
        }
    }
}

```

OUTPUT

```

ENTER START
0
ENTER END
400
0
1
2
3
4
5
6
7
8
9
153
370
371

```

CONCEPT

An Armstrong number is a number where:

sum of each digit raised to the power of total digits = original number

Formula:

$(d1^n + d2^n + d3^n + \dots) = \text{number}$

Where:

- $d1, d2, \dots$ = digits
- n = number of digits

Example 1: 153

153 has 3 digits

Calculation:

$$\begin{aligned} &1^3 + 5^3 + 3^3 \\ &= 1 + 125 + 27 \\ &= 153 \end{aligned}$$

Since result = original number
→ 153 is Armstrong